

TVEITSUND, Norway

WMO: 014550

Lat: 59.027N

Lon: 8.521E

Elev: 827

StdP: 14.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.5	8.6	0.8	5.9	6.9	5.0	7.3	10.2	15.2	40.6	12.7	38.6	1.8	30	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	77.6	61.2	74.1	59.8	70.7	58.6	63.6	71.7	62.1	69.7	60.6	67.8	4.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.9	82.2	65.9	59.2	77.4	64.3	57.7	73.3	62.9	29.3	72.3	28.2	69.5	27.2	67.8	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	9.8	8.5	-0.2	82.5	7.2	3.2	-5.3	84.9	-9.6	86.7	-13.6	88.6	-18.8	90.9		
(5)			1.5	66.0	7.2	2.1	-3.7	67.5	-8.0	68.7	-12.0	69.8	-17.3	71.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	28.2	27.3	32.5	40.6	48.8	56.3	60.4	58.7	52.0	43.0	35.5	30.0
	DBStd	13.47	8.66	9.31	7.70	5.60	6.29	4.73	4.72	4.28	4.65	5.76	6.86	8.77
	HDD50	3559	677	636	543	285	99	6	0	1	32	226	434	620
	HDD65	8099	1142	1056	1008	732	501	263	159	199	389	681	884	1085
	CDD50	957	0	0	0	3	63	195	322	271	93	10	0	0
	CDD65	23	0	0	0	0	0	3	16	4	0	0	0	0
	CDH74	311	0	0	0	0	20	67	188	34	2	0	0	0
	CDH80	34	0	0	0	0	1	6	26	1	0	0	0	0
Wind	WSAvg	3.8	3.8	3.8	3.9	3.7	3.8	3.9	3.6	3.5	3.6	3.7	3.8	4.0
Precipitation	PrecAvg	42.3	3.9	2.4	2.2	2.4	2.9	3.4	3.9	4.8	4.3	4.7	4.3	3.1
	PrecMax	64.2	9.5	7.6	5.6	6.3	8.0	7.3	9.3	9.0	14.0	11.9	18.2	6.0
	PrecMin	32.1	0.6	0.4	0.4	0.2	0.2	0.4	0.6	0.4	1.6	0.8	1.0	0.6
	PrecStd	7.4	2.2	1.7	1.3	1.5	1.9	1.7	2.0	2.2	2.6	2.7	3.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.1	48.6	55.6	65.1	76.7	80.2	83.3	78.0	69.5	60.1	54.6	51.0
		MCWB	42.5	42.1	44.5	50.1	58.4	60.1	62.4	63.2	58.2	53.3	48.6	45.8
	2%	DB	44.5	45.2	50.1	58.7	70.0	75.0	78.9	74.6	65.6	56.3	51.4	47.4
		MCWB	40.1	40.0	42.0	47.0	54.7	58.1	61.5	62.3	56.1	51.4	47.8	43.6
	5%	DB	42.0	42.6	46.6	54.0	65.3	70.7	75.3	71.1	62.7	54.3	48.5	44.8
		MCWB	38.5	38.0	39.8	44.8	53.0	56.4	60.7	60.8	55.2	50.5	45.0	41.3
	10%	DB	39.7	39.9	43.6	50.6	60.9	67.3	71.2	68.0	60.1	52.3	45.6	42.1
		MCWB	36.7	36.4	38.0	42.9	50.7	55.1	59.5	58.6	54.1	49.0	42.7	39.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	43.6	45.9	52.0	59.8	62.5	65.9	65.7	60.7	56.0	51.1	46.6
		MCDB	46.5	47.8	53.6	60.0	72.2	73.0	76.2	73.2	65.5	57.8	53.0	49.4
	2%	WB	41.2	40.8	43.0	48.5	56.9	60.5	63.8	63.8	58.5	53.2	48.2	44.4
		MCDB	44.0	44.2	48.8	55.8	66.4	70.1	73.2	70.6	62.0	55.1	50.7	47.1
	5%	WB	39.1	38.7	40.6	46.3	54.3	58.6	62.2	62.3	57.1	51.6	45.7	41.9
		MCDB	41.4	41.7	45.4	52.8	62.4	67.5	70.7	68.7	60.4	53.5	48.2	44.3
	10%	WB	37.1	36.7	38.8	44.3	52.1	56.8	60.8	60.7	55.5	49.5	43.2	39.3
		MCDB	39.6	39.6	42.9	49.3	59.2	64.4	68.6	66.3	58.8	51.8	45.4	41.7
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.9	14.0	15.9	17.6	17.8	16.8	15.5	13.1	10.9	8.7	9.1
		MCDBR	11.7	13.0	19.3	23.9	26.5	25.7	25.0	20.5	18.6	13.3	11.9	12.1
		MCWBR	9.5	9.4	11.9	13.6	13.3	11.9	10.9	9.9	10.9	10.0	10.2	10.1
	5% WB	MCDBR	10.6	11.6	16.8	21.3	22.4	21.1	19.6	16.8	13.7	10.4	10.6	10.8
MCWBR		9.3	9.1	11.1	13.0	12.0	11.1	9.8	9.1	9.5	9.0	9.7	9.7	
Clear-Sky Solar Irradiance	taub	0.255	0.291	0.316	0.359	0.356	0.343	0.363	0.359	0.332	0.315	0.275	0.232	
	taud	2.227	2.308	2.332	2.332	2.431	2.486	2.443	2.447	2.520	2.494	2.326	2.156	
	Ebn at Noon	182	234	262	268	278	283	274	266	257	226	174	147	
	Edh at Noon	14	21	28	33	32	31	32	30	23	18	13	10	
All-Sky Solar Radiation	RadAvg	106	302	771	1254	1600	1831	1633	1274	827	381	141	72	
	RadStd	23	64	68	178	190	140	187	88	108	68	36	9	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page